Discrete Recursive Models for Continuous Robotic Control: Applying TRM and HRM to PushT

VLA-HRM Project

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Abstract

We investigate applying discrete recursive reasoning models — Tiny Recursive Models (TRM) and Hierarchical Reasoning Models (HRM) — to the PushT robotic manipulation benchmark from Diffusion Policy. These models, originally designed for discrete puzzles (Sudoku, Mazes, ARC-AGI), perform iterative refinement through recursive computation. We bridge the continuous-discrete gap through careful tokenization of action and observation spaces, and introduce dimension-aware positional encoding to preserve the structure of flattened multi-dimensional sequences. We conduct extensive experiments on tokenization strategies, positional encodings, model sizes, and recursion depths.

1 Introduction

Diffusion Policy [1] has shown strong performance on robotic manipulation tasks by formulating action prediction as a denoising diffusion process. However, recent work on discrete recursive models — TRM [2] and HRM [3] — has demonstrated remarkable reasoning capabilities on tasks like Sudoku and ARC-AGI with very compact models (7–27M parameters).

A natural question arises: *Can the recursive reasoning mechanism of these models be leveraged for robotic control?* The iterative refinement process in TRM/HRM is analogous to the denoising process in diffusion models — both progressively transform an initial estimate into a final prediction. However, the key challenge is that robotic actions are continuous, while TRM/HRM operate on discrete tokens.

1.1 Contributions

- Adaptation of TRM and HRM architectures for continuous robotic control via discrete tokenization
- Dimension-aware positional encoding for structured multi-dimensional sequences
- Extensive ablation studies on tokenization, positional encoding, model size, and recursion depth
- Comparison against autoregressive GPT baseline and Diffusion Policy

2 Background

2.1 PushT Task

PushT is a planar pushing task where a circular agent must push a T-shaped block to match a target pose. The state space is 5-dimensional: agent position (x_a, y_a), block position (x_b, y_b), and block angle θ_b. Actions are 2-dimensional target positions (x_t, y_t) for PD control. The evaluation metric is coverage (intersection area / goal area), with success at 95% coverage.

The dataset contains 206 human demonstrations with 25,650 total timesteps. Following Diffusion Policy conventions, we use an observation horizon of $T_o = 2$ and a prediction horizon of $T_a = 16$.

2.2 Recursive Reasoning Models

TRM (Tiny Recursive Model) uses a single reasoning module that is recursively applied in a two-level loop:

- **L-level**: Low-level computation cycles (L_{cycles} iterations)
- **H-level**: High-level refinement cycles (H_{cycles} iterations)

The same weight-shared module processes both levels, alternating between updating the low-level state z_L (with high-level input injection) and the high-level state z_H (with low-level input injection).

HRM (Hierarchical Reasoning Model) extends TRM with *separate* modules for the H-level and L-level:

- H-level module: slow, abstract planning with its own transformer layers
- L-level module: fast, detailed computation with its own transformer layers

Both models only backpropagate through the final reasoning cycle, making training memory-efficient despite deep effective computation.

3 Method

3.1 Tokenization

We discretize continuous observations and actions into a vocabulary of V bins. We explore two strategies:

Uniform Binning divides each dimension's range $[\min, \max]$ into V equal-width bins:

$$\text{token}(x) = \left\lfloor \frac{x - x_{\min}}{x_{\max} - x_{\min}} \cdot V \right\rfloor, \quad \hat{x}(\text{token}) = x_{\min} + (\text{token} + 0.5) \cdot \frac{x_{\max} - x_{\min}}{V} \tag{1}$$

Mu-law Encoding applies logarithmic compression before binning, giving finer resolution near the center of the range:

$$f(x) = \text{sign}(x) \cdot \frac{\ln(1 + \mu|x|)}{\ln(1 + \mu)} \tag{2}$$

3.2 Sequence Structure

We flatten observations and actions into a single token sequence:

$$\underbrace{s_1^0, s_2^0, s_3^0, s_4^0, s_5^0, s_1^1, \ldots, s_5^1}_{\text{observations: } T_o \times 5 = 10}, \underbrace{a_1^0, a_2^0, a_1^1, \ldots, a_2^{15}}_{\text{actions: } T_a \times 2 = 32}$$

During training, all tokens are provided (teacher forcing). During inference, action tokens are replaced with a learned mask embedding.

3.3 Dimension-Aware Positional Encoding

Standard positional encodings treat the flattened sequence as a 1D list, losing the multi-dimensional structure. We propose encoding three aspects:

$$\mathrm{PE}(i) = \mathrm{TimeEmb}(t_i) + \mathrm{DimEmb}(d_i) + \mathrm{TypeEmb}(\tau_i) \tag{3}$$

where t_i is the timestep index, d_i is the dimension index within that timestep, and $\tau_i \in \{0, 1\}$ indicates observation vs. action.

4 Experimental Setup

- **Hardware**: NVIDIA A100-PCIE-40GB
- **Training**: 500 epochs, batch size 256, AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$)
- **Learning rate**: 10^{-4} with linear warmup (10 epochs) and cosine decay
- **Evaluation**: 50 episodes every 50 epochs
- **Logging**: Weights & Biases

5 Results

5.1 Baseline Comparison

Table 1: Baseline results on PushT. Diffusion Policy score from original paper.

Model	Params	Val Loss	Val Acc	Mean Score	Max Score
Diffusion Policy (DDPM)	2.5M	—	—	0.75	—
TRM V1 (tokenized, h=256)	1.4M	0.001	100.0%	0.061	0.067
HRM V1 (tokenized, h=256)	2.5M	0.001	100.0%	0.045	0.059
TRM V2 (cont. obs, h=256)	1.5M	3.12	16.5%	0.107	0.154
HRM V2 (cont. obs, h=256)	2.8M	3.22	15.5%	0.093	0.152
TRM V3 (regression, h=256)	1.5M	0.047	—	0.170	0.270
HRM V3 (regression, h=256)	**2.8M**	**0.035**	—	**0.236**	**0.344**
TRM V3 large (h=512)	7.5M	0.066	—	0.140	0.207
TRM V3 deep (H5L6, ah=2)	2.2M	0.111	—	0.091	0.143

5.2 Key Findings

Discrete tokenization fails for control. V1 models achieve 100% token prediction accuracy on training data but only 4–6% mean coverage in evaluation. This is because: (1) small quantization errors compound over the 16-step action horizon; (2) the model memorizes exact training sequences rather than learning a generalizable policy; (3) any deviation from training observations leads to out-of-distribution states.

HRM outperforms TRM. The hierarchical separation of H-level (planning) and L-level (execution) modules in HRM consistently outperforms the weight-shared TRM. At epoch 150, HRM achieves 23.6% mean score vs. TRM's 17.0%. This suggests that separate modules can specialize: H-level for trajectory planning and L-level for precise action computation.

Regression output is essential. Moving from classification (cross-entropy over bins) to regression (MSE on continuous actions) improved scores by $3-4\times$. Continuous outputs avoid quantization errors and allow smoother action trajectories.

Observation noise augmentation prevents overfitting. Adding Gaussian noise ($\sigma = 2.0$) to observations during training significantly improved generalization to unseen initial states.

6 Analysis

6.1 Recursion as Iterative Refinement

The recursive reasoning mechanism in TRM/HRM is analogous to the denoising process in diffusion models:

- **Diffusion**: iteratively denoise a random Gaussian sample toward an action sequence
- **TRM/HRM**: iteratively refine latent states z_H, z_L to produce better action predictions

The key advantage is weight sharing across all refinement steps, yielding compact models (2.8M params) compared to diffusion models (2.5M+ for state-based Diffusion Policy).

6.2 Gap to Diffusion Policy

Our best model (HRM V3, 0.34 max score) still significantly underperforms Diffusion Policy (0.75 mean score). Potential reasons:

- Diffusion Policy uses iterative denoising with hundreds of steps; we use 3–5 recursion cycles
- Diffusion Policy's action chunk prediction naturally handles multi-modality
- Our evaluation uses randomized initial states which may differ substantially from training distribution
- Training data is small (206 demonstrations) — more data augmentation may help

7 Conclusion

We demonstrated that recursive reasoning models (TRM/HRM) can be adapted for continuous robotic control, with the key insight that continuous observation encoding and regression outputs are essential. The hierarchical HRM architecture outperforms the flat TRM, achieving 34.4% maximum coverage on PushT. While still below Diffusion Policy's 75%, this represents a promising direction for compact, parameter-efficient robotic policies based on iterative refinement.

Future work: (1) more recursion cycles with gradient checkpointing, (2) multi-modal output heads (GMM), (3) larger training datasets, (4) temporal action ensemble from multiple recursion depths.

References

[1] Chi, C., et al. "Diffusion Policy: Visuomotor Policy Learning via Action Diffusion." RSS 2023.

[2] Samsung SAIL Montreal. "Tiny Recursive Models." 2024.

[3] Sapient Inc. "Hierarchical Reasoning Model." 2024.